

MVP — Personal Item Inventory

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1. Problem

1.1. Problem statement

People who own durable items that are used infrequently often rely on memory or informal lists to keep track of what they own and where those items are stored. As a result, they may purchase items they already own, spend significant time searching for rarely used belongings, or keep items for years simply because they are unsure whether they may need them in the future. These behaviors can contribute to unnecessary spending, wasted time, and accumulation of unused possessions

1.2. Evidence

Five people were interviewed, and the following questions were asked:

- How often does this happen?
- What types of objects are involved?
- How much time do you spend looking for things?
- Do you maintain any kind of list?
- Have you ever thrown something away because you didn't know whether you might need it later?
- Have you kept something for years because you're afraid you'll need it someday?

1.3. Key findings

The main findings are:

Evidence area	Findings
Duplicate purchases	4 out of 5 people reported having bought something they already owned at least once.
Frequency	In general, it does not occur very frequently; responses range from occasionally to approximately once a year.
Types of objects	The items mainly include small or low-cost items: food/snacks, drinks, stationery, books, tools, electronics, kitchen items, spare parts, clothing.
Searching for objects	The time it takes varies considerably. Some people hardly look at all because they remember where they put things; others report taking anywhere from 30 minutes to over an hour in certain situations.
Lists / tracking	There are mixed behaviors: some people use lists, notes or memory; others do not maintain any inventory.
Discarding items	Several people have thrown away items and later regretted it or wondered whether they should have kept them.
Keeping “just in case”	4 out of 5 show some behavior of keeping things because they might need them later, although they are not necessarily frequently used objects.
Location problem	The problem is stronger for items that are used infrequently and stored away, rather than everyday objects.

1.4. Workaround

Currently, users employ the following solutions to the problem at hand:

- Search closets, garage, storage rooms.
- Keep lists in notebooks.
- Rely on memory.
- Ask another family member.
- Simply buy another item when they can't find the first one.

2. Target User

2.1. Primary target user

The primary target users are people who own a significant number of durable items that they use infrequently and rely mainly on memory or informal lists to keep track of them. They may own tools, electronics, spare parts, games, household items, or other belongings that are stored away and are not part of their daily routine.

2.2. Possible segments

- Tool/equipment owners: tools, spare parts, electronics, equipment.
- Hobbyists: board games, art suppliers, sports equipment, DIY materials, electronics
- General household users

3. Goals

The MVP should help users:

- Find an item when they need it.
- Remember whether they own an item and where it is stored.
- Decide whether to keep, discard, sell, or donate an item.
- Avoid unnecessary purchase.

4. Value Proposition

These goals are supported by four main value areas:

4.1. Ownership Memory: Avoid duplicate purchases

Help users remember what they own so they can check whether they already have an item before buying another one.

4.2. Location & Retrieval: Find rarely used items quickly

Help users remember where an item is stored and retrieve it without spending unnecessary time searching.

4.3. Decision Support: Make better decisions about what to do with an item

Help users decide whether to keep, discard, sell, or donate belongings based on relevant information.

4.4. Low-Maintenance Inventory: Track only what is difficult or costly to manage

The system does not require users to create a complete inventory. Users can choose which belongings are worth registering.

Key hypothesis

Users will perceive enough value from tracking selected belongings to justify the effort required to maintain the inventory.

5. MVP Features

5.1. Add / Register Item

The user can register a selected belonging.

Minimum information

- Item name
- Category
- Storage location

Optional information

- Photo
- Description
- Quantity

5.2. Storage Location

The user can assign a location to an item.

Example:

Home → Garage → Shelf 2 → Box A

The exact depth of this hierarchy has yet to be validated.

5.3. Search Inventory

The user can search for an item.

Example:

“Hammer”

Resultado:

Garage → Shelf 2 → Box A

The system returns matching items and their locations.

5.4. Item Details

The user can view the information associated with an item.

Example:

- Cordless Hammer
- Category: Tools
- Location: Garage → Shelf 2
- Status: Keep
- Last updated: September 2026

The exact fields should remain minimal.

5.5. Item Status / Decision

The user can assign a simple status:

- Keep
- Sell
- Donate
- Discard

This directly supports the decision-support value proposition.

The system can simply help the user record and review their decision.

5.6. Decision-Relevant Information

To make the four statuses meaningful, the user may need a small amount of additional information about an item.

Possible MVP fields:

- How often it is used
- Approximate replacement cost
- Last used
- Condition

5.7. Edit Item

The user can update information when something changes.

For example:

- **Garage → Storage Room**

This is important because an inventory becomes useless if its information becomes outdated.

6. MVP Scope

The MVP will provide a simple personal inventory that helps users remember what they own, find rarely used belongings, and make decisions about what to do with them. The user decides which belongings are worth registering.

6.1. Inventory Management

- Add / register an item
- Edit an item
- Delete an item
- View item details

6.2. Ownership Memory

- Store information about registered belongings
- Search the inventory to check whether an item is already owned

6.3. Location & Retrieval

- Assign a storage location to an item
- View the item's storage location
- Search for an item and retrieve its location

6.4. Decision Support

- Assign one of four statuses:
 - Keep
 - Sell
 - Donate
 - Discard
- Store a limited amount of relevant information that can help the user make this decision

6.5. Low-Maintenance Inventory

- Keep item registration simple
- Minimize the amount of information required
- Make updates easy when item information changes

6.6. Out of Scope

- Automatic duplicate detection
- AI-generated recommendations
- Marketplace integration
- Automatic selling or donation
- Social or sharing features
- Household collaboration

7. Functional Requirements

7.1. Inventory Management

FR-01 — Add Item

The system shall allow the user to create an item record.

The item shall contain, at minimum:

- Name
- Category
- Storage location

Optional information may include:

- Photo
- Description
- Quantity
- Decision-relevant information

FR-02 — View Item

The system shall allow the user to view the information associated with a registered item.

FR-03 — Edit Item

The system shall allow the user to modify an item's information, including its storage location and status.

FR-04 — Delete Item

The system shall allow the user to remove an item from the inventory.

7.2. Ownership Memory

FR-05 — Search Item

The system shall allow the user to search the inventory for a registered item.

FR-06 — Identify Ownership

The system shall allow the user to determine whether a matching item exists in the inventory by searching for the item's name.

7.3. Location & Retrieval

FR-07 — Store Location

The system shall allow the user to associate a storage location with an item.

FR-08 — Display Location

The system shall display the storage location when the user views an item.

7.4. Decision Support

FR-09 — Assign Item Status

The system shall allow the user to assign one of the following statuses to an item:

Keep / Sell / Donate / Discard

FR-10 — Store Decision-Relevant Information

The system shall allow the user to store a limited set of information that may support their decision about an item.

7.5. Decision Support

FR-11 — Manage Categories

The system shall allow the user to create, edit, activate, and deactivate item categories.

FR-12 — Manage Storage Locations

The system shall allow the user to create, edit, activate, and deactivate Level 1 and Level 2 storage locations.

FR-13 — Manage Units of Measure

The system shall allow the user to create, edit, activate, and deactivate units of measure.

8. Acceptance Criteria

FR-01 — Add Item

Given the user has entered the required item information, when the user submits the item, then the system creates and stores the item successfully.

The system should prevent saving when required information is missing.

FR-03 — Edit Item

Given an existing item, when the user changes its information and saves it, then the updated information is stored and displayed.

FR-04 — Delete Item

Given an existing item, when the user deletes the item, then the item is removed from the inventory and no longer appears in the inventory or search results.

FR-05 — Search Item

Given an item exists in the inventory, when the user searches using its name, then the item appears in the search results.

The system should also provide understandable feedback when there are no matching items.

FR-06 — Identify Ownership

Given the user is considering purchasing an item, when they search the inventory, then they can determine whether a matching item is already registered.

FR-08 — Display Location

Given an item has a stored location, when the user opens the item, then the system displays its storage location clearly.

FR-09 — Assign Item Status

Given the user is viewing an item, when they choose a status, then the system stores and displays the selected status.

Valid statuses are: Keep / Sell / Donate / Discard

FR-11 — Manage Categories

Given the user is in Settings, when the user creates a category, then the category becomes available for item selection.

Given an active category, when the user deactivates it, then the category is no longer available for new item selection.

FR-12 — Manage Storage Locations

Given an active Level 1 location, when the user creates a Level 2 location under it, then the Level 2 location becomes available for item selection.

FR-13 — Manage Units of Measure

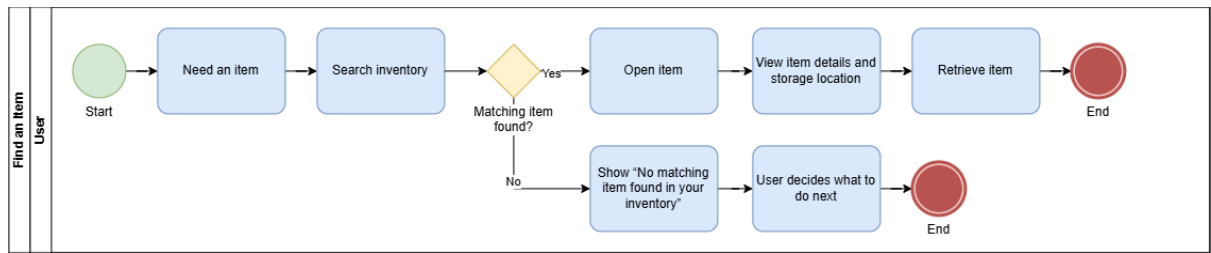
Given the user creates a unit of measure, then the unit becomes available when entering item quantity.

9. User Flow / UX

The UX should focus on the main user situations identified in the problem and interviews.

9.1. Main Flow — Find an Item

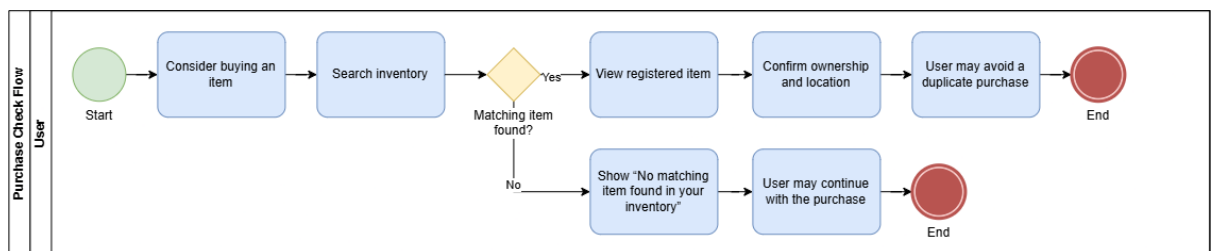
Goal: Find a registered item and retrieve its stored location



The system should make the item's location easy to understand and avoid requiring unnecessary navigation.

9.2. Purchase Check Flow

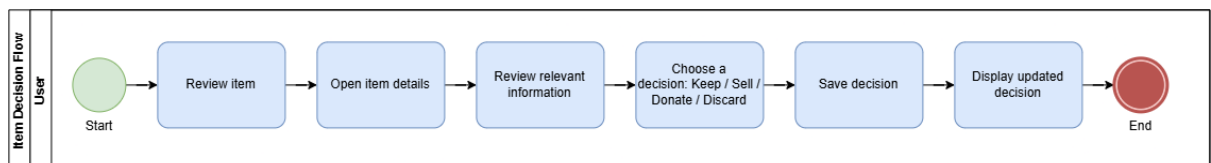
Goal: Check whether the user already owns an item before purchasing it.



The system should not imply that no search result means the user does not own the item. It only means that no matching registered item was found.

9.3. Item Decision Flow

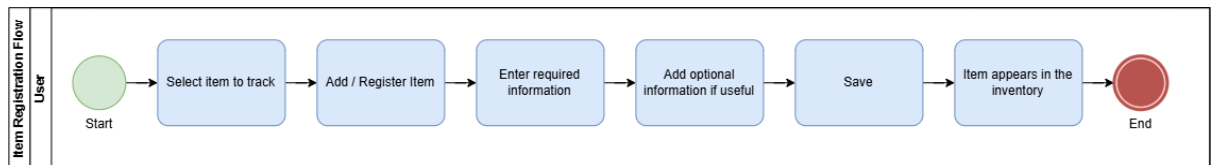
Goal: Help the user review an item and record a decision.



For the MVP, the system does not recommend a decision. The user makes the decision, while the system stores and presents the relevant information.

9.4. Item Registration Flow

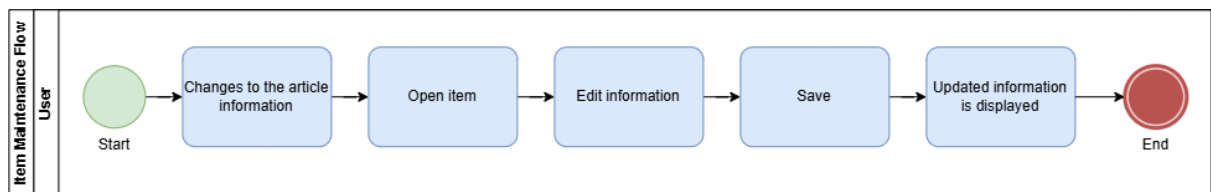
Goal: Register a selected belonging with a small amount of information.



The user decides which belongings are worth registering. The system does not require a complete inventory.

9.5. Item Maintenance Flow

Goal: Keep inventory information reasonably current.



Examples include changing the storage location or updating the item's decision status.

9.6. Important UX States

The main flows should account for:

- Empty inventory
- No search results
- One matching result
- Multiple matching results
- Item without optional information
- Item without a decision/status
- Editing an existing item
- Successful save
- Missing required information
- Save or system error

The UX should make the next action clear in each state and provide understandable feedback when an action cannot be completed.

10. Risks & Open Questions

10.1. Risk 1 — Maintenance Effort

Risk

Users may find registration and maintenance more effortful than the value they receive.

Fallback

Minimize required fields, keep the basic item record simple, and make updates quick.

10.2. Risk 2 — Data Accuracy

Risk

Storage locations, item information, or decisions may become outdated, reducing the usefulness of the inventory.

Fallback

Make editing quick and visible so users can update information when something changes.

10.3. Risk 3 — Decision Support Scope

Risk

The decision-support feature could expand into AI recommendations, pricing, resale integration, or complex scoring.

Fallback

For the MVP, the user makes the decision. The system only stores and presents relevant information and the selected decision.

10.4. Risk 4 — Storage Model Complexity

Risk

A detailed physical-storage hierarchy could become unnecessarily complex before the actual need is validated.

Fallback

Start with a simple storage location field and refine the model only if evidence requires more structure.

10.5. Risk 5 — Low Frequency of the Problem

Risk

The problem does not occur very frequently for many users. Duplicate purchases were reported as occasional events, in some cases approximately once a year.

Users may therefore not perceive enough value from maintaining an inventory to justify the effort.

Fallback

Keep the MVP low-maintenance and focus on selected belongings that are difficult to remember, rarely used, or time-consuming to find. Validate whether the perceived benefit is sufficient to justify the maintenance effort.

10.6. Risk 6 — Low Frequency of the Problem

Risk

The interviews provide evidence of keeping items “just in case” and regretting some discarded items, but they do not yet establish that users need a dedicated decision-support feature.

Fallback

Keep the feature simple: allow users to review a small amount of relevant information and record Keep / Sell / Donate / Discard. Validate whether users find this useful.

10.7. Open Questions

- How much information are users willing to enter when registering an item?
- Is avoiding occasional duplicate purchases valuable enough to justify maintaining the inventory?
- How much time are users willing to spend registering and updating an item?
- Which information is actually useful when deciding Keep / Sell / Donate / Discard?
- How should the system handle similar item names, such as multiple versions of the same type of item?
- Is a simple location field sufficient, or do users need a hierarchical location such as Home → Garage → Shelf → Box?
- How often do users need to update item information?
- Will users continue maintaining the inventory over time?
- Does the proposed low-maintenance approach provide enough value without requiring users to track everything they own?

11. Success Criteria

The MVP should be considered successful if it demonstrates value across the four proposed value propositions.

11.1. Ownership Memory

Users can use the inventory to answer:

“Do I already own this?”

Evidence of success:

- Users can find registered items through search.

- Users can understand whether a matching item exists.
- Users report that this information can help them avoid unnecessary purchases.

11.2. Location & Retrieval

Users can answer:

“Where is it?”

Evidence of success:

- Users can find an item's stored location.
- The location information is understandable.
- Users can complete the retrieval flow without unnecessary confusion.

11.3. Decision Support

Users can answer:

“What should I do with this item?”

Evidence of success:

- Users can review relevant information.
- Users can assign Keep / Sell / Donate / Discard.
- Users consider the stored information useful when making the decision.

11.4. Low-Maintenance Inventory

Users can maintain their inventory without excessive effort.

Evidence of success:

- Users can add an item with only a small amount of information.
- Users can update information easily.
- Users do not need to create a complete inventory to obtain value.
- Users perceive the maintenance effort as reasonable compared with the benefit.